

Nikhilesh Vijay Chirutala

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 2.8+ years of experience building and maintaining production applications across AI, trading, CRM, and cloud platforms. Strong experience with Python, FastAPI, REST APIs, MongoDB, AWS, and cloud application development. Hands-on experience implementing Retrieval-Augmented Generation (RAG) and integrating LLM capabilities for contextual information retrieval. Experienced in data processing, backend service development, API integration, testing, troubleshooting, and production deployments, with a strong foundation in software design, problem solving, and application architecture.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, API Development, Backend Automation, Authentication, RBAC, ACL

Data & Databases: MongoDB, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2

Integrations & Testing: Twilio, GitHub API, Postman, Jest, Unit Testing, Git, GitHub

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

- Develop Python backend services and automation for a production cloud contact center platform, supporting application workflows, data processing, and cloud-based operations.
- Design and maintain Python and FastAPI services and REST APIs, translating business requirements into reliable backend functionality and integrations.
- Develop AWS-based solutions for application data and storage workflows, applying validation and error handling to improve operational reliability.
- Investigate production issues, perform root-cause analysis, implement bug fixes and feature enhancements, and validate changes before release.

Key Projects:

CONVOSO — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, REST APIs

- Developed backend services supporting lead routing, call recording management, data processing, and cloud storage workflows.
- Designed recording-recovery workflows processing 10,000+ recordings per month and restoring missing or inconsistent recording data.
- Implemented S3 storage validation and metadata synchronization to maintain consistency between application records and cloud objects.
- Built automation to identify, recover, organize, and synchronize recordings across storage systems, reducing manual intervention.
- Added validation and error handling for failed recording operations and unexpected storage states to improve recovery reliability.
- Traced recording pipeline failures across backend services, cloud storage, and metadata workflows to identify and implement fixes.
- Monitored recording and storage workflows in production, validating processing outcomes and addressing failures to maintain reliable operations.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Built production backend services, REST APIs, and application workflows using Python, FastAPI, and MongoDB across multiple applications.
- Developed and integrated an AI-powered trading platform, including RAG-based conversational retrieval and backend data-processing workflows.
- Implemented authentication, RBAC, and ACL-based authorization and integrated third-party APIs including Twilio and GitHub API.
- Supported testing, debugging, production issue resolution, CI/CD, and application deployments across AWS and Vercel.

Key Projects:

AI Bull — AI-Powered Trading Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, and back-testing.
- Implemented a Retrieval-Augmented Generation (RAG) chatbot that retrieves relevant financial and application-specific information to generate contextual responses.
- Integrated LLM capabilities into application workflows to provide conversational access to relevant financial information.
- Built FastAPI/Python backend services for financial data processing, business logic, and AI application workflows.
- Developed configurable screening, filtering, and back-testing capabilities for stocks and mutual funds.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested APIs and workflows with Postman and resolved bugs across CRM functionality and production workflows.

GharPe — Real Estate Management Platform

React.js, FastAPI, Python, MongoDB, REST APIs

- Developed the React.js-based admin system for managing real estate projects and controlling their visibility and availability on the website.
- Built project management workflows allowing administrators to create, update, configure, and manage projects displayed across the platform.
- Developed lead management functionality to track and manage customer inquiries generated through the platform.
- Built builder management features for maintaining builder information and associating builders with their respective projects.
- Developed and integrated FastAPI/Python REST APIs supporting administrative workflows, project configuration, and data operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

Contact Management App — Contacts Storage Application

React.js, JavaScript, Netlify

- Developed a React-based application for creating, viewing, updating, and managing contact information.
- Built reusable UI components and deployed the application on Netlify.

CERTIFICATIONS

- Agile Methodology Virtual Experience Program — Cognizant (Oct 2022)
- Python Programming Online Training — APSSDC (May–Jun 2021)

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur